Week Report 6

Wildcards.

* Wildcard

- A star alone matches anything and nothing and matches any number of characters.
- Examples:
  - lists all the text files in a directory
    - `ls *.txt`
  - list all the files that start with the word file
    - `ls file*`
  - Copy all the mp4 files
    - `cp Downloads/*.mp4 ~/Videos/Movies/`

? Wildcard

- The ? wildcard metacharacter matches precisely one character.
- Examples:
  - lists all the files in the current directory
    - `ls`
  - list all the hidden files in the current directory
    - `ls ./.??`
  - list all the files that have a 3 letter file extension.
    - `ls *.???`

[] wildcard

- The brackets wildcard match a single character in a range.
  - Examples:
    - to match all files that have a vowel after letter f:
      - `ls f[aeiou]*`
    - to match all files whose name has at least one number:
      - `ls[0-9]*`
    - to match all files that have a range of letters after f:
      - `ls f[a-z]*`

Brace expansion

- Brace expansion {} is not a wildcard but another feature of bash that allows you to generate arbitrary strings to use with commands.
  - Examples:
    - To create a whole directory structure in a single command:
      - `mkdir -p music/{jazz,rock}/{mp3files,videos,oggfiles}/new{1..3}`
    - Remove multiple files in a single directory:
      - `rm -r {dir1,dir2,dir3,file.txt,file.py}`

- To create a N number of files use:
  - `touch website{1..5}.html`
  - `touch file{A..Z}.txt`

Practice

- Practice 5:

```
rsarikaya@rsarikaya-VirtualBox:~$ curl -s https://cis106.com/assets/wildcardprac
tice.sh | bash
--2022-11-16 14:35:45--  https://cis106.com/assets/logo100x100.png
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3520 (3.4K) [image/png]
Saving to: ‘logo100x100.png’

logo100x100.png     100%[===================>]   3.44K  --.-KB/s    in 0s

2022-11-16 14:35:45 (10.9 MB/s) - ‘logo100x100.png’ saved [3520/3520]

rsarikaya@rsarikaya-VirtualBox:~$ ls wildcardpractice/*random*
wildcardpractice/_random.csv     wildcardpractice/randomtext.txt
wildcardpractice/random_text.md
rsarikaya@rsarikaya-VirtualBox:~$ ls wildcardpractice/*.txt wildcardpractice/*.c
sv
wildcardpractice/78food.txt      wildcardpractice/Sample1234.csv
wildcardpractice/_random.csv     wildcardpractice/sample123.csv
wildcardpractice/randomtext.txt
rsarikaya@rsarikaya-VirtualBox:~$ cd wildcardpractice/
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls random*.md
random_text.md
```

```
rsarikaya@rsarikaya-VirtualBox:~$ ls wildcardpractice/*random*
wildcardpractice/_random.csv     wildcardpractice/randomtext.txt
wildcardpractice/random_text.md
rsarikaya@rsarikaya-VirtualBox:~$ ls wildcardpractice/*.txt wildcardpractice/*.c
sv
wildcardpractice/78food.txt      wildcardpractice/Sample1234.csv
wildcardpractice/_random.csv     wildcardpractice/sample123.csv
wildcardpractice/randomtext.txt
rsarikaya@rsarikaya-VirtualBox:~$ cd wildcardpractice/
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls random*.md
random_text.md
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls *sample* *.md
random_text.md  sample1.doc  sample99.docx        sample.pdf
sample123.csv   sample1.xls  samplejsonfile.json  samplesite.html
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$
```

- Practice 6:

```
rsarikaya@rsarikaya-VirtualBox:~$ cd wildcardpractice/
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls *.???
2020-08-27-capture.raw  baik.pdf   bk.pdf    file.log          sample1.doc
2022-02-02-camera.jpg   baik.xml   bk.xml    logo100x100.png   sample1.xls
78food.txt              beatk.exe  book.exe  _random.csv       sample.pdf
allfiles.lst            beatk.pdf  book.pdf  randomtext.txt    Sample.pdf
Allfiles.lst            beatk.xml  book.xml  Sample1234.csv
baik.exe                bk.exe     fail.log  sample123.csv
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls .??*
.bash_history  .hidden1.py   .hidden2.py
.hidden1.doc   .hidden2.doc  .hiddenFile.txt
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls b??.???
ls: cannot access 'b??.???': No such file or directory
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls b??*.???
baik.exe  baik.xml   beatk.pdf  book.exe  book.xml
baik.pdf  beatk.exe  beatk.xml  book.pdf
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls sample?.*
sample1.doc  sample1.xls
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls.??*.??
ls.??*.??: command not found
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls .??*.??
.hidden1.py  .hidden2.py
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$
```

- Practice 7:

```
rsarikaya@rsarikaya-VirtualBox:~$ cd wildcardpractice/
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [a-z]*.???
allfiles.lst  beatk.exe  bk.pdf    book.xml         randomtext.txt  sample.pdf
baik.exe      beatk.pdf  bk.xml    fail.log         sample123.csv
baik.pdf      beatk.xml  book.exe  file.log         sample1.doc
baik.xml      bk.exe     book.pdf  logo100x100.png  sample1.xls
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [A-Z]*[0-9]*
Sample1234.csv
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [0-9][0-9][0-9]*
2020-08-27-capture.raw  2021-04-10-screenshot.jpeg  2022-02-02-camera.jpg
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [0-9][0-9][0-9][0-9]*
2020-08-27-capture.raw  2021-04-10-screenshot.jpeg  2022-02-02-camera.jpg
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [!AEIOUaeiou]
ls: cannot access '[!AEIOUaeiou]': No such file or directory
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [!AEIOUaeiou]*
2020-08-27-capture.raw      beatk.pdf  fail.log         sample1.doc
2021-04-10-screenshot.jpeg  beatk.xml  file.log         sample1.xls
2022-02-02-camera.jpg       bk.exe     logo100x100.png  sample99.docx
78food.txt                  bk.pdf     _random.csv      samplejsonfile.json
baik.exe                    bk.xml     random_text.md   sample.pdf
baik.pdf                    book.exe   randomtext.txt   Sample.pdf
baik.xml                    book.pdf   Sample1234.csv   samplesite.html
beatk.exe                   book.xml   sample123.csv
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [[:punt:]]
```

```
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [[:punct:]]
ls: cannot access '[[:punct:]]': No such file or directory
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$ ls [[:punct:]]*
_random.csv
rsarikaya@rsarikaya-VirtualBox:~/wildcardpractice$
```